

Exp 8.cpp Exp9.cpp Exp10.cpp Exp11.cpp Exp12.cpp Exp13.cpp [*] Exp14.cpp Exp15.cpp Exp16.cpp Exp17.cpp Exp18.cpp Exp19.cpp Exp20.cpp

```
1 #include <stdio.h>
2 #include <pthread.h>
3 #include <semaphore.h>
4 sem_t mutex, writeBlock;
5 int data = 0, readersCount = 0;
6
7 void *reader(void *arg) {
8     int i = 0;
9     while (i < 10) {
10         sem_wait(&mutex);
11         readersCount++;
12         if (readersCount == 1) {
13             sem_wait(&writeBlock);
14         }
15         sem_post(&mutex);
16
17         // Reading operation
18         printf("Reader reads data: %d\n", data);
19
20         sem_wait(&mutex);
21         readersCount--;
22         if (readersCount == 0) {
23             sem_post(&writeBlock);
24         }
25         sem_post(&mutex);
26     }
27 }
```

sources Compile Log Debug Find Results Close

Compilation results...

```
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- Errors: 0
- Warnings: 0
- Output Filename: C:\AADHITHYA\Exp20.exe
- Output Size: 186.2158203125 KiB
- Compilation Time: 1.22s
```

globals

Exp 8.cpp Exp9.cpp Exp10.cpp Exp11.cpp Exp12.cpp Exp13.cpp [*] Exp14.cpp Exp15.cpp Exp16.cpp Exp17.cpp Exp18.cpp Exp19.cpp Exp20.cpp

```
25     sem_post(&mutex);
26     i++;
27 }
28 return NULL; // FIXED: Added missing return statement for void* function
29 }
30
31 void *writer(void *arg) {
32     int i = 0;
33     while (i < 10) {
34         sem_wait(&writeBlock);
35
36         // Writing operation
37         data++;
38         printf("Writer writes data: %d\n", data);
39
40         sem_post(&writeBlock);
41         i++;
42     }
43     return NULL; // FIXED: Added missing return statement for void* function
44 }
45
46 int main() {
47     pthread_t readers, writers;
48
49     sem_init(&mutex, 0, 1);
```

resources Compile Log Debug Find Results Close

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(globals)

Exp 8.cppExp 9.cppExp 10.cppExp 11.cppExp 12.cppExp 13.cpp[*] Exp 14.cppExp 15.cppExp 16.cppExp 17.cppExp 18.cppExp 19.cppExp 20.cpp

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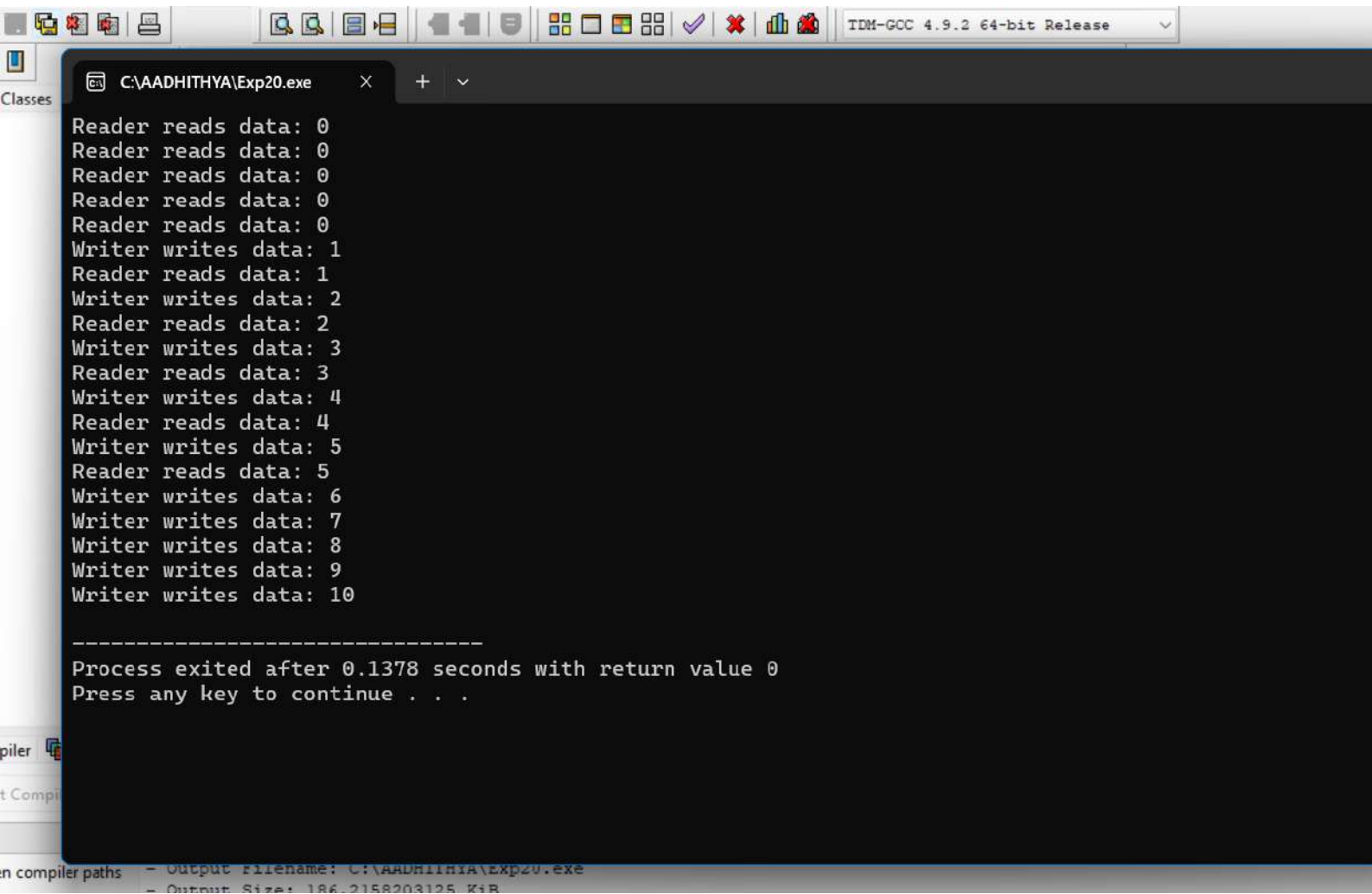
```
sem_post(&writeBlock);  
i++;  
}  
return NULL; // FIXED: Added missing return statement for void* function  
}  
  
int main() {  
    pthread_t readers, writers;  
  
    sem_init(&mutex, 0, 1);  
    sem_init(&writeBlock, 0, 1);  
  
    // FIXED: Cleaned up spacing for split lines  
    pthread_create(&readers, NULL, reader, NULL);  
    pthread_create(&writers, NULL, writer, NULL);  
  
    pthread_join(readers, NULL);  
    pthread_join(writers, NULL);  
  
    sem_destroy(&mutex);  
    sem_destroy(&writeBlock);  
  
    return 0;  
}
```

resources

Compile LogDebugFind ResultsClose

Compilation results...

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```
Reader reads data: 0
Reader reads data: 0
Reader reads data: 0
Reader reads data: 0
Reader reads data: 0
Writer writes data: 1
Reader reads data: 1
Writer writes data: 2
Reader reads data: 2
Writer writes data: 3
Reader reads data: 3
Writer writes data: 4
Reader reads data: 4
Writer writes data: 5
Reader reads data: 5
Writer writes data: 6
Writer writes data: 7
Writer writes data: 8
Writer writes data: 9
Writer writes data: 10

-----
Process exited after 0.1378 seconds with return value 0
Press any key to continue . . .
```

Compiler paths - Output filename: C:\AADHITHYA\Exp20.exe
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